Hydrobiologia (2026) 853:145–166
https://doi.org/10.1007/s10750-025-05921-2

PRIMARY RESEARCH PAPER

Effects of flooding on aquatic macroinvertebrate assemblage structure on a connected and disconnected regulated river floodplain

Patricia L. S. Wooden · William E. Kelso · Michael D. Kaller

Received: 16 August 2024 / Revised: 23 April 2025 / Accepted: 5 June 2025 / Published online: 16 June 2025
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2025

Abstract The Atchafalaya River Basin is the largest continuously forested floodplain in the United States. This ecosystem has been heavily modified for flood control and human use, often with minimal study into ecological effects on resident biota. As part of modifications, sections of the historic floodplain have been disconnected from the annual flood pulse via an extensive levee system. We examined the influence of the annual flood pulse on aquatic macroinvertebrate assemblages, with an emphasis on Hemiptera, by sampling a connected floodplain and a disconnected floodplain lake across two flood pulses of varying magnitude in 2018 and 2019. On the connected floodplain, flooding-related changes to temperature, dissolved oxygen, and water depth were important factors differentiating early flood and late-flood assemblages. Disconnected floodplain insect assemblages greatly differed from connected floodplain by 50% of taxa, lower richness, and lower densities of 21 of 24 taxa shared between both floodplains. Flooding magnitude between years delayed development of the late-flood assemblage later into the flood pulse. Results suggest, in terms of aquatic insect abundance and composition, aerial dispersal from adjacent (< 100 m) floodplain areas cannot replace loss of direct river connectivity. Further, restoration of riverine flooding is important for re-establishing aquatic macroinvertebrate assemblages in disconnected floodplains.

Keywords Hemiptera · Mississippi river · Seasonal flood pulse · Macroinvertebrates · Atchafalaya river

Handling editor: Hu He

P. L. S. Wooden
Department of Entomology, Louisiana State University Agricultural Center, Baton Rouge, LA, USA

P. L. S. Wooden (✉)
Department of Agricultural Science and Plant Protection, Mississippi State University, Starkville, MS, USA
e-mail: benthicbugologist@gmail.com

W. E. Kelso · M. D. Kaller
School of Renewable Natural Resources, Louisiana State University Agricultural Center, Baton Rouge, LA, USA

Introduction

River floodplains provide hydrologically complex environments with considerable spatiotemporal heterogeneity in physiology and chemistry related to sub-annual or annual flood pulses (e.g., Sabo et al., 1999a, b; Starr et al., 2014). Although effects of inundation on floodplain biota vary with hydrogeomorphology and geography (Gallardo et al., 2014), flood duration and magnitude (Chessman and Hardwick, 2014; Chanut et al., 2018), and river-floodplain connectivity (Quinn et al., 2000; Zilli et al., 2008; Leigh and Sheldon, 2009), flooding strongly influences the abundance and species composition of floodplain organisms (McInerney et al., 2017; Stoffels et al., 2022;

Munyai et al., 2023). For macroinvertebrates, the abundance, distribution, and dynamics of both benthic (Zilli et al., 2008; Dube et al., 2019; Willbanks and Mullis, 2022) and epiphytic (e.g., Hemiptera: Nepidae, Diptera: Culicidae) (Boulton and Lloyd, 1991; Poi de Neiff and Carignan, 1997; Obolewski et al., 2014) taxa are strongly tied to the magnitude and duration of flooding, which is related to the spatial arrangement and connectivity of waterbodies on the floodplain landscape (e.g., Boulton and Lloyd, 1991; Starr et al., 2014; Dube et al., 2019). Because macroinvertebrate assemblage composition varies across gradients in the timing, magnitude, and duration of river inputs, these assemblages have been used to characterize floodplain function and provide metrics (e.g., Paillex et al., 2007, 2013) for the design and assessment of floodplain restoration projects (Besacier-Monbertrand et al., 2014; Paillex et al., 2015; Jennings et al., 2023).

The Atchafalaya River Basin (ARB), a 4,900-km^2 forested floodplain wetland of the Atchafalaya River, is characterized by a diversity of aquatic habitats, including shallow lakes, bayous, and canals excavated for oil and gas development (Piazza, 2014). The floodplain annually receives Atchafalaya River water beginning in late fall/winter, with floodwaters receding in early to late summer. The flood pulse plays a dominant role in determining ARB physicochemistry (Sabo et al., 1999a, b), which is reflected in abundance and distribution patterns of recreationally and commercially important fishes and invertebrates (Alford and Walker, 2013; Vargas-Lopez et al., 2020). Increasing temperatures and decomposition of organic matter in floodplain aquatic habitats (exacerbated by detritus from invasive water hyacinth *Eichhornia crassipes* and *Hydrilla verticillata*) during the latter stages of the flood pulse result in spatiotemporally extensive hypoxia [dissolved oxygen (DO) levels < 2.0 mg/l] that significantly alters abundances of resident organisms (Fontenot et al., 2001; Pasco et al., 2016; Kroes et al., 2022). The ARB is constrained by east and west guide levees that were constructed in the late 1940s, resulting in isolated lakes outside the levees that are no longer influenced by the Atchafalaya River pulse, although they still exhibit minor precipitation-related changes in lake levels and summer hypoxia (Gallardo et al., 2023).

Studies of macroinvertebrate ecology in the ARB floodplain have focused primarily on macrophyte-mediated effects on macroinvertebrate distribution and abundance. These studies have examined macroinvertebrate responses to the spatial relationships between macrophyte beds and water inputs from the Atchafalaya River, macrophyte species composition and architecture, or with variability in the annual hydrologic cycle. None of the previous studies have investigated disconnection from the Atchafalaya River's annual flood pulse and usually focused on the post flood period (e.g., Colon-Gaud et al., 2004; Fisher et al., 2012). Generally, closer spatial proximity to flood water inputs with associated higher dissolved oxygen and increased nutrients resulted in greater macroinvertebrate densities (Battle and Mihuc, 2000; Mihuc et al., 2000, 2002; Fisher et al., 2012). Greater relative composition of native plants, such as coontail (*Ceratophyllum demersum*), compared with invasive hydrilla (*Hydrilla verticillata*) increased macroinvertebrate abundance and diversity due to substantial differences in dissolved oxygen levels between the native and invasive dominated macrophyte beds (Colon-Gaud et al., 2004). However, invasive hydrilla still does provide habitat and refuge from predation (Fisher et al., 2012). The annual hydrologic cycle influences wild native crayfish (*Procambarus clarkii*) with a similar positive association between annual flooding magnitude and harvest noted in both fishery-independent (Pollard et al., 1983; Bonvillain et al., 2012, 2014) and fishery-dependent data (Alford and Walker, 2013; Vargas-Lopez et al., 2020). Taken together, the positive influence of better oxygenated and nutrient rich water from the Atchafalaya River has indirect (i.e., the promotion of beneficial macrophyte beds from nutrients and compensation for decomposition from greater dissolved oxygen) and direct (i.e., greater levels of dissolved oxygen) to macroinvertebrates. Thus, the influence of complete disconnection upon vegetation-dwelling macroinvertebrates is likely negative viewed from these direct and indirect pathways. However, this has not been investigated directly, despite a large extent of disconnected floodplain. Further, little research has addressed the potential role of annual ARB water level fluctuations on floodplain macroinvertebrate assemblage composition and dynamics.

The spatial proximity of flood-pulse influenced (connected) and non-flood-pulse influenced (disconnected) waterbodies inside and outside the ARB guide levees, respectively, provides a unique opportunity to examine

potential effects of water level fluctuations on resident biota. We were particularly interested in abundance patterns of aquatic and semi-aquatic Hemiptera, as these insects are often integral parts of lotic and lentic food webs, serving as predators and prey to both vertebrates and invertebrates (Murdoch et al., 1984; Polhemus, 2008; Foltz and Dodson, 2009). Hemipterans have been found to be common in wetlands across the world, with elevated abundances in systems with prolonged flood events (Batzer and Ruhí, 2013; Turić et al., 2015). Not limited by mandibles, the piercing-sucking hemipterans are able to feed on prey much larger than themselves and can be dominant invertebrate predators in many aquatic habitats (Matheson, 1907; Wilson, 1958; Peckarsky et al., 1990; Papacek, 2001). Hemipterans readily colonize new habitats as adults, including temporarily flooded areas (Tronstad et al., 2007), and many taxa have evolved cyclic colonization and unique migration habits in response to hydrologically dynamic wetlands (Batzer and Ruhí, 2013). Moreover, Ruhí and Batzer, (2014) suggested a Mollusca, Hemiptera, and Coleoptera (MHC) metric be established for assessment of wetland invertebrate assemblages, as these more tolerant taxa, particularly belostomatid, corixid, gerrid, and notonectid hemipterans (Golladay et al., 1997; Tronstad et al., 2007; Batzer and Ruhí, 2013), are often abundant in dynamic wetlands systems.

In this study, we were interested in relationships among macroinvertebrates, vegetation, and water quality for samples collected in the ARB during the 2018 and 2019 flood pulses, particularly given the substantial differences in pulse magnitude and duration between the two years. We hypothesized that gradients in flow, temperature, and DO would result in distinct assemblages during the course of the flood pulse. We also hypothesized that a former floodplain currently disconnected from the ARB flood pulse would support an altered abundance and distribution of resident macroinvertebrates relative to a connected floodplain.

Methods and materials

Sampling locations

The ARB is the largest continuous forested floodplain in the United States (Fig 1). For evaluating response to the flood pulse and to serve as the connected floodplain, we selected Buffalo Cove (BC), in the southwestern portion of the ARB. BC has been the focus of a restoration project since 2004, and although substantial information exists on the water quality, hydrology, fishes, and macrophytes (e.g., Lambou and Hern, 1983; Trumbo et al., 2016; Kroes et al., 2022), no invertebrates other than red swamp crayfish (*Procambarus clarkii*; Vargas-Lopez et al., 2020) have been investigated in aquatic habitats of the BC floodplain.

For comparison to BC, we selected the Lake Fausse Pointe (LFP) region, which, prior to levee construction, was historically part of the larger Atchafalaya River floodplain and received floodwaters during elevated river stages. This same design was employed by Gallardo et al., (2023) to examine flood pulse impacts on periphyton composition and dynamics. LFP was separated from the rest of the ARB by a levee that was completed in 1958. The lake shares a 26-km border with the ARB, and distances between connected (BC) and disconnected (LFP) locations are often separated by less than 100 m of levee. Lakes Fausse Pointe and Dauterive (to the north) cover an area of 6,880 hectares and were historically natural lakes that drained into the ARB. With construction of the west guide levee, LFP now feeds drains into the Vermillion-Teche River watershed. Although hydrologic input and output are regulated by anthropogenic control structures and the system is regularly maintained for wildlife and vegetation control, the LFP system does experience precipitation-driven annual changes in water levels of ~1.2 m (Tye, 1986).

Sampling

Sampling was timed to coincide with floodplain inundation, with 2018 high water (river stage > 3 m at the Butte la Rose gauge LA-07381515) sampling conducted from March 7 to May 22, and low water (river stage < 3 m at the Butte la Rose gauge LA-07381515) sampling from June 25 to September 16. The magnitude of flooding in 2019 was unusually high and protracted, limiting sampling to only high water conditions (March 7 to August 2). Sampling sites were selected for their differences in hydrology and habitat characteristics. Site BC2 was characterized by dense accumulations of water hyacinth (*Pontederia crassipes*), and in previous years had become inaccessible by boat in late summer because of accumulations of

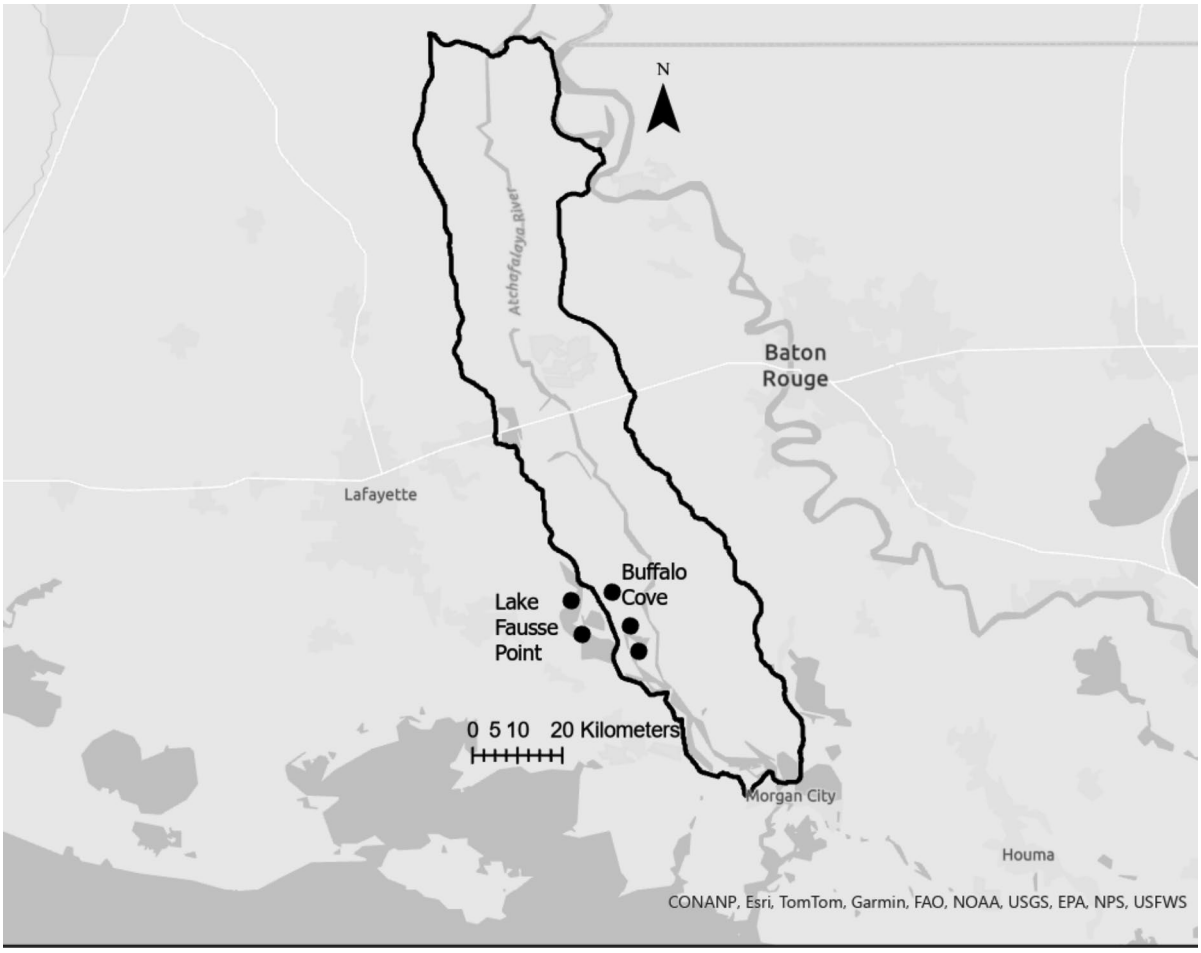

Fig 1 Sampling locations in the connected Buffalo Cove floodplain and disconnected Lake Fausse Pointe floodplain. Black boundary denotes the regulated Atchafalaya River Floodway System bounded by levees to the north, south, east, and west

dense water hyacinth mats. BC1 and BC3 contained small floating mats of water hyacinth, if present, but supported stands of both submerged and emergent macrophytes. In LFP, traps were placed in a windswept shoreline that supported rooted macrophytes and occasional mats of water hyacinth (LFP4), and in a protected area where floating water hyacinth and water lilies (Nymphaecae) formed dense mats (LFP5).

During each sampling event in 2018 and 2019, four to six submerged quatrefoil light traps (Secor et al., 1992) used as activity traps (Anderson et al., 2013) were placed adjacent to canal banks in BC along macrophyte beds in water depths of approximately 50 cm and at a minimum distance of 10 m apart resulting in two to three light traps on one side of the canal and two to three light traps on the other side of the canal. The number of light traps deployed depended on accessibility within the canal (e.g., whether it was possible to find a full 30 m of open water adjacent to the canals). Light traps were collected after approximately 24 h [7–11 sampling events (depending on site accessibility) × 3 sites × 4–6 light traps = 199 light traps deployed in BC]. During low water conditions in 2018, two sampling sites were added in LFP following the same light trap deployment procedure as in BC (3 sampling events × 2 sites × 6 traps = 36 light traps deployed in LFP; Fig 1). In 2019, unusually high precipitation associated with Hurricane Barry (11–15 July), Tropical Storm Imelda (17–19 September), and Tropical Storm Olga (22–25 October) extended high water past the normal early summer

flooding, resulting in lack of access due to the closing of Lake Fausse Point State Park. Light trap samples were preserved in 70% alcohol upon collection.

Not all light trap deployments collected organisms, but 164 samples from BCWMU and 24 samples from LFP yielded macroinvertebrates for data analyses. Water velocity was measured at each trap location, and a channel surface water quality measurement was taken with a portable water quality meter [specific conductance (mS), DO (mg/l), temperature (°C), pH and turbidity (NTUs)] when traps were deployed and retrieved (ProDSS Multiparameter Water Quality Meter, YSI Incorporated, Yellow Springs, OH). Coverage of aquatic vegetation also was estimated on a 0–100% scale at each light trap location. To evaluate sampling effectiveness of light traps, we collected a subsurface sweep net sample at the completion of light trap sampling if small rafts of water hyacinth were present. In 2018, a suitcase sample (Colon-Gaud and Kelso, 2003) was also taken to collect macroinvertebrates at the edge of intact water hyacinth mats that had developed in water shallow enough to use the sampler (wadeable). Suitcase samples were not taken during the 2019 flood pulse due to extremely high water levels that precluded effective sampling. Both suitcase and sweep samples were placed in plastic bags and frozen upon return to the lab. In the lab, insects and other arthropods were identified to family or order, and hemipterans were identified to the lowest practical taxonomic level with appropriate guides [Gonsoulin, 1973; Epler, 2006; Merritt et al., 2008 (most recent edition at time of study); Thorp and Rogers, 2011] and comparison with reference specimens in the Louisiana State Arthropod Museum.

Statistical analysis

Prior to analyses, the number of organisms per trap was standardized to a 24-h sample. A standardized river stage (SWL) was calculated by averaging the Atchafalaya River stage (ARS) at the Butte la Rose gauge for the entire sampling period and calculating the difference in river stage on each sampling date to the overall average. Statistical analyses first examined relationships between macroinvertebrates, water quality, and vegetation for the BC data. Subsequent analyses focused on an assessment of the effects of river connectivity on macroinvertebrate abundances (BC versus LFP) during low water when both areas were sampled in 2018. Finally, given that flood magnitude and duration differed substantially during the two sampling years, we assessed differences in macroinvertebrate densities in BC between 2018 and 2019.

Water quality variables (standardized water level, flow velocity, surface temperature, dissolved oxygen, and vegetation density) are inherently multivariate as they influence each other (e.g., the well-known relationship between temperature and dissolved oxygen) and thus differences in water quality between the connected and disconnected floodplain were analyzed by permutational multivariate analysis of variance (PERMANOVA with Euclidean distance) with date and site as fixed effects (Program R vers. 4.2.1, package vegan Oksanen et al., 2019; R Core Team, 2023). Final model was determined by examining homogeneity of dispersion (Anderson et al., 2006) and Dunn-Smyth residual outputs.

Analyses associating macroinvertebrate assemblage composition with water quality (SWL, water velocity, surface water temperature, DO, turbidity, and specific conductance) and vegetation density were based on assemblage measures: Hemiptera and arthropods (non-Hemiptera) per trap, rarefied Hemiptera and rarefied arthropod richness per trap, and Hemiptera and arthropod diversity (Shannon) per trap. These datasets were analyzed with generalized linear mixed models (GLMMs) and multivariate analyses. Diversity and rarefied richness were estimated at the genus level for rarefied Hemiptera richness and at the family level for arthropod richness following Hurlbert, (1971) and Heck et al., (1975; Program R vers. 4.2.1, package vegan, Oksanen et al., 2019; R Core Team, 2023). GLMMs included SWL, water velocity, surface temperature, dissolved oxygen, pH, turbidity, and vegetation density as fixed effects and date sampled as a random effect with a log link, negative binomial distribution and Laplace approximation (Program R vers. 4.2.1, package lme4, Bates et al., 2015; R Core Team, 2023). For multivariate analyses, we used permutational multivariate analysis of variance (PERMANOVA) and multivariate GLMMs with an appropriate error distribution (normal, Poisson), after comparing other canonical link and distribution combinations. For all multivariate analyses, SWL, surface temperature, DO, turbidity, vegetation density, date, and site were fixed effects (Program R vers. 4.2.1, package vegan Oksanen et al., 2019, package mvabund Wang et al., 2012; package ggplot

Wickham, 2016; package lattice Sarker, 2008; package RColorBrewer Neuwirth, 2022; package patchwork Pederson 2024; R Core Team, 2023).

We also used GLMMs to assess the effects of Atchafalaya River connectivity on assemblage measures. For these analyses, connectedness was a fixed effect and sampling site and date were random variables. For the multivariate assemblage analysis, PERMANOVA and multivariate GLMMs with an appropriate error distribution (normal, Poisson) included connectedness, site, and date as fixed effects.

Comparisons of BC macroinvertebrate assemblages between 2018 and 2019 also used GLMMs (rarefied arthropod richness, rarefied Hemiptera richness, and arthropods per trap) and linear mixed models (LMMs; Hemiptera per trap, Hemiptera diversity, and arthropod diversity) with year as fixed effect and site and date as random effects. LMMs were used when assumptions of linearity and normality were met. Similar to the above analyses, PERMANOVA and multivariate GLMMs with an appropriate error distribution (normal, Poisson) included year, site, and date as fixed effects. For all analyses described above, final models were determined by examining homogeneity of dispersion (Anderson et al., 2006) and Dunn-Smyth residual outputs. Influential taxa were identified by examining associations with each respective model.

Results

Comparison of catches from light traps and sweep/suitcase samples indicated light traps were an effective method for capturing the diversity of macroinvertebrates inhabiting these sampling sites, particularly for Ephemeroptera, Hemiptera, Trichoptera, Coleoptera, and Decapoda. With the exception of amphipods and isopods, light traps captured 3 to 5 times the number of organisms for all invertebrate orders encountered during the study (Table 1). The 2018 sampling period was a typical flood pulse year, characterized by a consistent spring decline in river stage (the floodplain becomes inundated at approximately 3 m), increasing temperatures through June, and relatively high mean DO levels throughout spring and late summer, with declining DO evident as the floodplain drained (Fig 2a). The 2019 flood pulse was atypical (Fig 2b), with high water levels extending through the year, and hypoxic DO levels evident during the summer. In contrast to BC, water levels were consistent throughout the sampling period in LFP, which exhibited high temperatures and DO levels during all sampling periods (Fig 2c). During summer 2018, turbidities ranged from 44 to 58 NTUs at all BC and LFP sites except BC2, and specific conductance was also minimally variable, ranging from 0.25 to 0.31 mS/cm at all sites (Table 2, Fig 3). Water velocities were slightly higher in BC than LFP, although all mean velocities were below 9 cm/sec, and surface temperatures were higher in LFP, which may have been related to reduced water movement. Vegetation densities were variable through time, but water hyacinth was the dominant species at all vegetated sites. Salvinia (*Salvinia sp.*), duckweed (*Lemna sp.*), alligator weed (*Alternanthera philoxeroides*), frogbit (*Limnobium laevigatum*) and coontail (*Ceratophyllum demersum*) were also common, especially at BC2, where dense vegetation and downed trees from Tropical Storm Barry (7/19/2019) prevented access and sampling in late July. Sampling in BC in 2018 occurred at slightly elevated river stages relative to the mean (SRS) compared to 2019, although absolute Atchafalaya River stages were much higher (ARS) during the extended, high magnitude flood pulse in 2019.

Table 1 Total count of macroinvertebrate taxa by order collected in Buffalo Cove and Lake Fausse Point with light traps and suitcase/sweep net samples in 2018 and 2019

Order	Light Trap	Suitcase and Sweep Samples
Ephemeroptera	326	4
Odonata	70	13
Hemiptera	2348	9
Neuroptera	6	2
Trichoptera	42	1
Lepidoptera	3	1
Coleoptera	382	70
Hymenoptera	8	4
Diptera	444	119
Polydesmida	2	0
Amphipoda	1750	2567
Arguloida	5	0
Decapoda	1131	27
Isopoda	38	38
Gastropoda	145	0

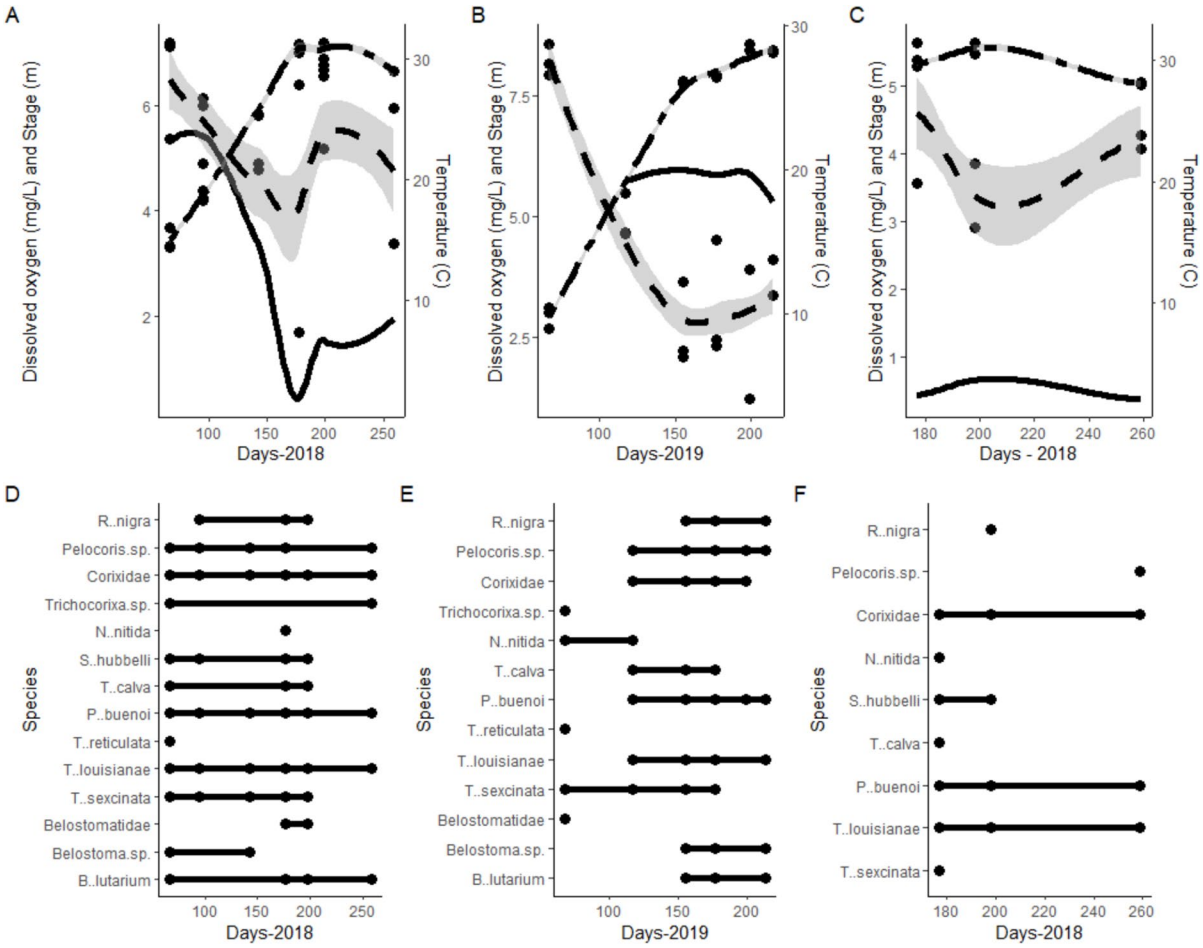

Fig 2 Relationships between river stage (solid black), dissolved oxygen (large dashes with shaded 95% confidence interval), and temperature (short dashes with shaded 95% confidence interval) for the connected Buffalo Cove in 2018 (**A**) and 2019 (**B**) and the disconnected Lake Fausse Pointe in 2018 (**C**) with temporal occurrence of Hemiptera species in Buffalo Cove in 2018 (**D**), Buffalo Cove in 2019 (**E**), and Lake Fausse Pointe (**F**), and log-transformed catch per unit effort in Buffalo Cove in 2018 (**G**), Buffalo Cove in 2019 (**H**), and Lake Fausse Pointe (**I**) illustrating the role of flooding with the development of earlier and later flood assemblages. R = *Ranatra*. S = *Sigura*. T = *Trichorixia*. P = *Pelocoris*. N = *Hesperocorixia*

Light trap samples contained 81 taxa of macroinvertebrates representing 50 families within 15 orders (Table 3), as well as occasional non-arthropod bycatch (*e.g.,* tadpoles and larval fish). Corixid hemipterans comprised a large proportion of the catch in both BC and LFP, whereas baetid mayflies, hydrophilid coleopterans, chironomid dipterans, amphipods, and palaemonid shrimp were also common. Eight families of aquatic Hemiptera (22 identified genera) were collected, many of which were newly associated with the ARB (Corixidae, Notonectidae, Veliidae, and Pleidae).

Environmental influences on macroinvertebrate assemblages

Although water quality measures were relatively similar within ARB and LFP (Table 4), the overall PERMANOVA indicated statistically significant differences between ARB and LFP ($F_{1,51} = 7.2$, $P < 0.01$), suggesting a complicated and interrelated response to disconnection. Differences were statistically significant for turbidity, SWL, specific conductance, temperature, and dissolved oxygen, in order of greatest to least difference. LFP exhibited lower SWL, specific

Table 2 Mean (±SE) water quality and vegetation data for sampling locations in Buffalo Cove (BC) and Lake Fausse Pointe (LFP) during 2018 and 2019. SRS and ARS are standardized and actual Atchafalaya River stage, respectively

Variable	BC1 2018	BC2 2018	BC3 2018	LFP4 2018	LFP5 2018	BC1 2019	BC2 2019	BC3 2019
SRS (m)	0.57 (0.30)	0.59 (0.25)	0.94 (0.27)			0.02 (0.04)	0.01 (0.05)	0.10 (0.04)
ARS (m)	– 0.46 (0.30)	– 0.45 (0.25)	– 0.10 (0.27)			4.00 (0.40)	3.96 (0.05)	4.24 (0.04)
Turbidity (NTU)	44.75 (7.84)	26.14 (1.32)	57.85 (9.82)	56.87 (11.06)	45.17 (5.43)	14.44 (1.68)	14.56 (1.44)	18.01 (1.07)
Flow (cm/sec)	4.86 (1.46)	7.79 (1.49)	8.73 (1.85)	2.75 (0.70)	1.75 (0.33)	8.61 (1.40)	4.03 (0.68)	9.22 (2.46)
Specific Conductance (mS)	0.31 (0.02)	0.31 (0.02)	0.29 (0.02)	0.27 (0.02)	0.25 (0.01)	0.31 (0.01)	0.31 (0.01)	0.33 (0.01)
Temperature (C^0)	23.63 (1.41)	23.92 (1.12)	21.82 (1.34)	29.83 (0.43)	29.4 (0.28)	22.93 (1.11)	21.86 (1.21)	23.23 (1.47)
Dissolved Oxygen	6.18 (0.16)	4.92 (0.27)	5.61 (0.29)	3.51 (0.14)	4.59 (0.23)	3.75 (0.40)	4.12 (0.40)	4.90 (0.32)
Vegetation coverage (%)	9.1 (2.9)	30.0 (7.4)	13.6 (4.4)	1.7 (1.7)	16.7 (10.1)	4.5 (2.4)	26.6 (6.7)	6.9 (3.0)

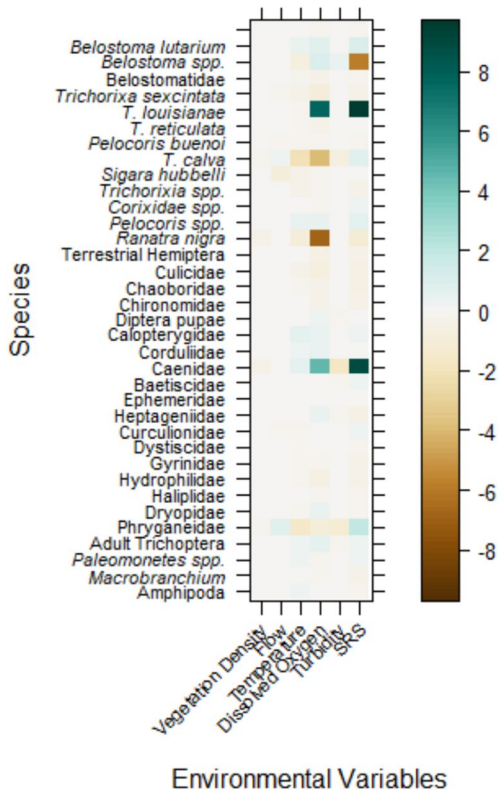

Fig 3 Magnitude (parameter estimates) of positive (green) and negative (brown) relationships between macroinvertebrates and environmental variables

conductance, and dissolved oxygen and higher turbidity and temperature.

Total arthropod rarefied richness was negatively influenced by water velocity, whereas Hemiptera per trap was negatively influenced by temperature, specific conductance and dissolved oxygen and was positively influenced by pH (Table 5). In contrast, arthropods per trap, rarefied Hemiptera diversity, and rarefied arthropod diversity were not significantly associated with any of the measured environmental variables. Multivariate analyses indicated SWL (Poisson multivariate GLMM), flow velocity (Poisson multivariate GLMM), temperature (Poisson multivariate GLMM), DO (Poisson multivariate GLMM), turbidity (Poisson multivariate GLMM), and vegetation density (Poisson multivariate GLMM) were all associated with densities of several macroinvertebrate taxa, although associations varied substantially within and among orders (Table 6). Several taxa, notably *Belostoma lutarium* (Hemiptera: Belostomatidae), *Trichorixa calva* (Hemiptera; Corixidae), Hepatageniidae (Ephemeroptera), and Drypopidae (Coleoptera), appeared to be "late flood" taxa, being positively associated with higher temperatures and lower DO levels. Calopterygidae and Corduliidae (Odonata) were also contained in this group and were also positively associated with higher plant abundances and lower turbidities, both of which characterize the ARB floodplain later in the year. Conversely, Belostomatidae (Hemiptera), *Trichorixa reticulata* (Hemiptera:

Table 3 Total count of macroinvertebrate taxa collected in light traps from Buffalo Cove (BC) and Lake Fausse Point (LFP) in 2018 and 2019. UID = unidentified

Order	Family	Taxon	BC 2018	BC 2019	LFP 2019
Ephemeroptera	Baetidae		26	158	0
	Baetiscidae		0	8	0
	Caenidae		7	87	15
	Ephemeridae		18	9	60
	Heptageniidae		1	10	0
	Leptophlebiidae		1	0	0
Odonata	Aeshnidae		2	0	0
	Calopterigidae		6	53	16
	Coenagrionidae		0	1	0
	Corduliidae		2	3	1
	Gomphidae		0	1	0
	Libellulidae		0	1	0
	Petaluridae		0	1	0
Hemiptera	Veliidae		0	2	0
	Gerridae	*Limnoporus canaliculatus*	0	1	0
	Nepidae	*Ranatra* sp.	0	0	4
		R. nigra	6	9	1
	Belostomatidae	*Belostoma* sp.	21	47	0
		B. lutarium	6	4	1
		B. fusciventra	0	4	0
	Corixidae	juveniles	163	1257	45
		Corsiella edulis	0	1	0
		Hesperocorixa nitida	0	8	1
		Palmacorixa buenoi	79	84	325
		Ramphocorixa accuminata	0	0	1
		Sigaria sp.	0	2	0
		S. hubbelli	4	0	5
		S. virginiensis	0	3	0
		Trichocorixa sp.	7	4	0
		T. calva	5	14	7
		T. louisianae	75	34	8
		T. reticulata	1	5	0
		T. sexcincta	24	26	6
	Naucoridae	*Pelocoris* sp.	25	17	1
	Notonectidae	*Notonecta* sp.	2	0	0
	Pleidae	*Neoplea sp.*	2	0	0
Neuroptera	UID		1	5	1
	Corydalidae		1	2	0
	Sisyridae		0	0	1
	Terrestrial		0	3	0
Trichoptera	UID		8	34	3
	Adult		0	27	0
	Hydropsychidae		4	0	0
	Hydroptilidae		0	2	0
	Leptoceridae		2	0	2
	Philopotamidae		2	0	0
	Phryganeidae		0	4	1

Table 3 (continued)

Order	Family	Taxon	BC 2018	BC 2019	LFP 2019
	Rhyacophilidae		0	1	0
Lepidoptera	UID		0	3	0
	Noctuidae		0	3	0
Order	Family	Taxon	BC 2018	BC 2019	LFP 2019
Coleoptera	UID		221	161	45
	Curculionidae		4	4	1
	Dryopidae		0	16	0
	Dytiscidae		11	38	1
	Gyrinidae		34	35	23
	Halipidae		19	26	3
	Hydrophilidae		150	42	17
	Noteridae		2	0	0
	Scirtidae		1	0	0
Hymenoptera	UID		5	3	0
	Brachonidae		0	3	0
	Formicidae		5	0	0
Diptera	UID		334	110	59
	Adult		128	50	48
	Ceratopogonidae		0	1	1
	Chaoboridae		50	15	2
	Chironomidae		148	42	8
	Culicidae		6	1	0
	Stratiomyidae		2	1	0
Polydesmida	Polydesmidae		1	1	0
Amphipoda	UID		504	1246	177
Arguloida	Argulidae		4	1	7
Decapoda	Cambaridae		10	1	0
	Palaemonidae		736	384	283
Isopoda	UID		3	35	0
Gastropoda	UID		6	139	3
		Totals	2885	2355	1183

Corixidae), Culicidae (Diptera), Baetidae and Ephemeridae (Ephemeroptera) were positively associated with lower temperatures and higher DO levels, which are characteristic of the early flood pulse. Many hemipteran taxa appeared to be positively associated with SWL, suggesting higher abundances during periods of elevated water levels in both spring and summer.

Connectivity

During summer low water in 2018 when both locations were sampled, a substantial proportion (50%) of the 50 taxa collected were exclusive to either Atchafalaya River-connected BC (21 taxa) or disconnected LFP (4 taxa; Table 7). Of the assemblage metrics tested, only rarefied arthropod richness differed between connected (BC) and unconnected (LFP) locations (GLMM, log link, negative binomial, likelihood ratio test $X^2 = 5.77$, $P = 0.02$), with BC habitats having a back-transformed 0.71 ($\pm$ 1.13SE) higher richness per trap than LFP. None of the other metrics, including Hemiptera per trap, arthropods per trap, Hemiptera rarefied richness (GLMMs, log link, negative binomial, all likelihood ratio test $X^2 < 1.5$, $P > 0.46$), or Hemiptera diversity or arthropod diversity (LMMs, $F_{1,17.1} < 1.03$, $P > 0.33$) differed between connected

Table 4 Water quality measures for sampling locations in Atchafalaya River Basin (ARB) and Lake Fausse Pointe (LFP). Values are mean ± Standard Error. Bolded variables were statistically significant in comparisons between the ARB and LFP

	ARB 1 2018	ARB2 2018	ARB 3 2018	LFP 4 2018	LFP 5 2018	ARB 1 2019	ARB 2 2019	ARB 3 2019
Standardized Water Level (m)	0.57 ± 0.30	0.59 ± 0.25	0.94 ± 0.27	− 0.39 ± 0.01	− 0.11 ± 0.01	0.02 ± 0.04	0.01 ± 0.05	0.10 ± 0.04
Average Water Level (m)	− 0.46 ± 0.30	− 0.45 ± 0.25	− 0.10 ± 0.27	− 1.16 ± 0.01	− 3.82 ± 0.01	4.00 ± 0.04	3.96 ± 0.05	4.24 ± 0.04
Turbidity (NTU)	44.75 ± 7.84	26.14 ± 1.32	57.85 ± 9.82	56.87 ± 11.06	45.17 ± 5.43	14.44 ± 1.68	14.56 ± 1.44	18.01 ± 1.07
Flow (cm/sec)	4.86 ± 1.46	7.79 ± 1.49	8.73 ± 1.85	2.75 ± 0.70	1.75 ± 0.33	8.61 ± 1.40	4.03 ± 0.68	9.22 ± 2.46
Specific Conductance (um/s)	0.31 ± 0.02	0.31 ± 0.02	0.29 ± 0.02	0.27 ± 0.02	0.25 ± 0.01	0.31 ± 0.01	0.31 ± 0.01	0.33 ± 0.01
Temperature (C^0)	23.63 ± 1.41	23.92 ± 1.12	21.82 ± 1.34	29.83 ± 0.43	29.4 ± 0.28	22.93 ± 1.11	21.86 ± 1.21	23.23 ± 1.47
Dissolved Oxygen (mg/L)	6.18 ± 0.164	4.92 ± 0.27	5.61 ± 0.29	3.51 ± 0.14	4.59 ± 0.23	3.75 ± 0.40	4.12 ± 0.40	4.90 ± 0.32

Table 5 Relationships between environmental variables and assemblage metrics

Metric	Parameter Estimate (+SE)	Test Statistic	*P*-value
Rarefied arthropod richness	Water Velocity β = − 0.13 (± 0.06)	LR χ^2 = 20.44	< 0.01
Hemiptera per trap	Temperature β = − 1.45 (± 0.53)	LR χ^2 = 18.34	0.02
	Specific Conductance β = − 2.51 (± 0.50)	LR χ^2 = 18.34	0.02
	Dissolved Oxygen β = − 2.79 (± 0.78)	LR χ^2 = 18.34	0.02
	pH β = 1.11 (± 0.30)	LR χ^2 = 18.34	0.02

LR Likelihood ratio

with unconnected locations. In contrast to the linear models, multivariate analyses detected a significant difference in the assemblage composition between locations (Poisson multivariate GLMM, Deviance = 823, $P < 0.01$; Table 8), with 9 of 10 hemipteran taxa and 12 of 14 non-hemipteran taxa positively associated with connection to the Atchafalaya River. Assemblage differences between BC and LFP were also evident based on presence/absence of taxa during 2018 (Fig 2D, F). Several hemipteran taxa such as Corixidae, *Pelecoris buenoi* (Naurocidae), *Sigaria hubbelli* (Corixidae), and *Trichorixa louisianae* (Corixidae) appeared to be common members of the macroinvertebrate assemblages in the two locations for various periods throughout 2018. In contrast, *Ranatra nigra* (Nepidae), *Pelocoris* sp. (Naucoridae), *Trichorixa* sp., *T. calva*, *T. sexcintata* (Corixidae), *Belostoma* sp., and *B. lutarium* (Belostomatidae) were either exclusively or much more frequently encountered in BC.

Table 6 Positive (+) and negative (−) associations (parameters > |1.0|; $P < 0.05$) of ARB macroinvertebrates with measured environmental variables

Taxa	Vegetation Density (%) D = 459, $P < 0.01$	Flow velocity (cm/s) D = 908, $P < 0.01$	Temperature (C) D = 1713, $P < 0.01$	DO (mg/L) D = 459, $P < 0.01$	Turbidity (NTU) D = 321, $P < 0.01$	SWL (m) D = 459, $P < 0.01$
Hemiptera						
Belostoma lutarium			+	−		+
Belostoma spp.			−			−
Belastomatidae		+	−	+		+
Trichorixa sexcintata				+		+
T. louisianae				−		+
T. reticulata		+	−	+		+
Pelocoris buenoi				−		−
T. calva		−	+	−		+
S. hubbelli			+			−
Trichorixa spp.			−	+		+
Corixidae spp.				+		+
Pelocoris spp.			+			+
Ranatra nigra			−	−		−
Terrestrial Hemiptera			+	+		+
Diptera						
Culicidae		−	−	+	−	+
Chaoboridae				+		
Chironomidae				+		−
Diptera pupae					−	+
Odonata						
Calopterygidae	+	−	+	−	−	−
Corduliidae	+	−	+	+	−	−
Ephemeroptera						
Caenidae			−	−		−
Baetiscidae				−		
Baetidae			−	+		+
Ephemeridae			−	+		−
Heptageniidae			+	−	+	+
Colepotera						
Curculionidae			+	+		+
Dytiscidae				+		
Gyrinidae					−	
Hydrophilidae				−		−
Haliplidae						−
Dryopidae			+	−		−
Trichoptera						
Phryganeidae				+		+
Adult Trichoptera			−	+		+
Decapoda						
Paleamonetes sp.			−			−

Table 6 (continued)

Taxa	Vegetation Density (%) D = 459, $P < 0.01$	Flow velocity (cm/s) D = 908, $P < 0.01$	Temperature (C) D = 1713, $P < 0.01$	DO (mg/L) D = 459, $P < 0.01$	Turbidity (NTU) D = 321, $P < 0.01$	SWL (m) D = 459, $P < 0.01$
Macrobranchium sp.						–
Amphipoda				–	–	–

Terrestrial Hemiptera interact with aquatic macroinvertebrates in shallow wetlands, like these floodplains, therefore, they were included. *DO* Dissolved Oxygen, *SRS* Standardized Water Level, *D* Deviance

Table 7 Macroinvertebrates associated with sample sites that were either connected (ARB) or disconnected (LFP) from the Atchafalaya River in 2018

Connected Only	Disconnected Only
Hemiptera	
Unidentified immature Belostomatidae	Trichocorixia sp.
Trichocorixa sexcinata	
T. calva	
Ramphocorixa accuminata	
Hesperocorixa nitida	
Neoplea sp.	
Diptera	
Culicidae	
Chaoboridae	
Ceratopogonidae	
Odonata	
Ephemeroptera	Aeshnidae (Odonata)
Heptageniidae	
Leptophlebiidae	
Coleoptera	
Curculionidae	
Terrestrial	
Trichoptera	
Philoptamidae	
Phryganeidae	Leptoceridae (Trichoptera)
Megaloptera	
Corydalidae	
Sialidae	
Neuroptera	
Sisyridae	
Terrestria Neuroptera	
Isopoda	
Decapoda	
	Cambarellus peur

Table 8 Positive (+) and negative (-) associations (parameters > |1.0|) of macroinvertebrates with Atchafalaya River flooding in the ARB

Taxa	Association with River Flooding
Hemiptera	
Belostoma lutarium	+
Belostomatidae spp.	+
Pelocoris buenoi	+
Pelocoris spp.	–
Coridixidae spp.	+
T. louisianae	+
T. calva	+
T. sexinata1	+
S. hubbelli	+
Ranatra nigra	+
Diptera	
Culicidae	+
Chaoboridae	+
Chironomidae	–
Diptera pupae	+
Odonata	
Calopterygidae	–
Ephemeroptera	
Baetidae	+
Caenidae	+
Ephemeridae	+
Coleoptera	
Dytiscidae	+
Gyrinidae	+
Hydrophilidae	+
Haliplidae	+
Decapoda	
Paleamonetes spp.	+
Amphipoda	+

Normal (2018) and high (2019) water years

Macroinvertebrate assemblages in BC appeared to be very similar in 2018 and 2019 despite substantial differences in the timing, duration, and magnitude of the two flood pulses. Specifically, we did not detect any differences in arthropod richness (GLMM, log link, negative binomial, likelihood ratio test $X^2 = 0.12$, $P = 0.73$), total arthropods per trap (GLMM, log link, negative binomial, likelihood ratio test $X^2 = 0.07$, $P = 0.79$), total arthropod diversity per trap (LMM, $F_{1,111} = 0.50$, $P = 0.48$), Hemiptera richness (GLMM, log link, negative binomial, likelihood ratio test $X^2 = 2.48$, $P = 0.12$), Hemiptera per trap (LMM, $F_{1,111} = 0.50$, $P = 0.48$), or Hemiptera diversity per trap (LMM, $F_{1,111} = 3.33$, $P = 0.07$) between 2018 and 2019. However, several taxa, notably *Pelecoris* sp. (Naucoridae), Corixidae (*Trichorixa* sp., *T. calva*, *P. buenoi*, *T. louisianae*) *Belostoma* sp., and *B. lutarium* (Belostomatidae) appeared later in 2019 collections (Fig 2D, E).

Discussion & conclusions

Although we focused on Hemiptera, the annual river flood pulse was an important structuring factor across the macroinvertebrate assemblages examined, as demonstrated by direct comparisons of flooding-present and flooding-absent sites. Specifically, flooding had a direct effect on richness and densities of arthropods and Hemiptera and shaped early and late-flood assemblage compositions. Although within dispersal range, disconnection resulted in a 50% difference in taxa (21 unique to connected sites and only 4 unique to disconnected sites), greatly decreased overall hemipteran richness, and lower densities for 21 of 24 taxa common to both floodplains. Finally, regardless of higher or lower flood magnitude, hemipterans and other arthropods responded positively to flooding, with higher flood magnitude shifting in timing of development of the later flood assemblage. Therefore, similar to the findings of Gallardo et al., (2023) regarding floodplain periphyton assemblages, the annual flood pulse is a significant influence on invertebrate assemblage dynamics in this system.

Although invertebrate richness and diversity at the sampling sites were not strongly associated with most of the measured water quality variables, water level, water temperature, and dissolved oxygen were influential for several taxa, including multiple Hemiptera species. Our analyses included univariate descriptors of the arthropods (e.g., richness) and multivariate approaches to describe taxonomic composition. These analyses offered complementary insights into factors that enhance or suppress richness and shape assemblage composition. By adopting modern multivariate approaches, these analyses accounted for arthropod interactions among themselves and identified environmental characteristics that led to changes beyond the naturally occurring interactions among arthropods. Specifically, multivariate analyses described the development of early and late-flood pulse assemblages, similar to patterns described by Arrington and Winemiller (2006), Starr et al., (2014), Batzer et al., (2016, 2018), Dube et al., (2019), and Willbanks and Mullis, (2022). Given the Atchafalaya River flood pulse spans winter, spring, and summer seasons, separating phenological differences from flood progression is not wholly possible in a study of this duration. However, in multi-year studies, water level, water temperature, and dissolved oxygen in the Atchafalaya River were directly influenced by flooding, more so than season (e.., Sabo et al., 1999a,b; Kaller et al., 2011, 2015; Pasco et al., 2016; Kroes et al., 2022), suggesting the early and late season assemblages reflect connectivity and annual flooding. The positive relationship between insects and standardized water level has been shown previously, where abundance was positively associated with high fluctuations (Turić et al. 2015), and positive and negative associations between temperature and dissolved oxygen with aquatic arthropods in floodplains is well described (Gallardo et al., 2008, 2014; Batzer et al., 2016). A majority of species sampled were associated with higher dissolved oxygen and lower temperatures, typical of the earlier part of the flood pulse. For most insects, many of whom breathe dissolved oxygen from the water column, this relationship was expected. For hemipterans, which breathe atmospheric oxygen, this may have been a consequence of other flood factors, such as increased habitat for prey. Dissolved oxygen and temperature have an inverse relationship in ARB floodplains because of organic decomposition during low water in the hottest months of the year (Ford and Nyman, 2011; Kaller et al., 2015; Pasco et al., 2016; Kroes et al., 2022; Gallardo et al., 2023), and this may be the primarily driver of

the disappearance of the early flood pulse assemblage and replacement with the later flood pulse assemblage, which may have been more tolerant of lower dissolved oxygen and higher temperatures. Surprisingly, vegetation was not a strong indicator of arthropod diversity. The two taxa that did respond to vegetation were both odonates, which climb up vascular plants to emerge and are sit-and-wait predators of littoral areas (Corbet, 1999; Balzan, 2012). The lack of importance of aquatic vegetation as a driver of invertebrate assemblages in the ARB may be related to increases in vegetation density during the hot, low water season (Kaller et al., 2015; Vargas-Lopez et al., 2020), and thus, is a co-occurring factor along with changes in water level, temperature, and dissolved oxygen later in the flood (Turić et al., 2015).

Whereas we recognize that not all environmental factors were considered, many were beyond the scope of this work. Substrate and sediment characteristics are very important for benthic organisms (Shuman et al., 2020), but Hemiptera are not known to inhabit the benthos, relying on atmospheric oxygen. Similarly, aquatic hemipterans are rarely sprawlers or subject to burial by fine sediment inputs, which is detrimental to many other macroinvertebrates (Jones et al., 2012). Nutrient input data has historically been approximated for the Atchafalaya River (Robertson & Saad, 2021) but is lacking for the largely privately managed LFP and difficult to approximate for specific localities within the sprawling, seasonally inundated BC. Additionally, effects of excess nutrient loads have been shown to foster low dissolved oxygen (see below), a challenge already present in wetlands that is not negatively associated with aquatic Hemiptera (Spieles & Mitsch, 2000), but has not been shown to negatively impact wetland shredders and microbial communities (Batzer et al., 2004; Wissinger et al., 2021).

It is likely that climate change will exacerbate future differences in diversity and productivity of ARB insect assemblages relative to aquatic habitats outside the guide levees. Although both systems will be exposed to similar increases in temperature (ameliorated somewhat in the ARB by river water influence during the early flood pulse), land use change (Turner, 2022) and precipitation increases throughout the Mississippi River watershed (Lewis et al., 2023) will result in elevated river discharge, greater inputs of allochthonous and autochthonous organic material and nutrients (e.g., Gordon et al., 2020), and extended periods of flooding during river pulses relative to LFP and other systems separated from the flood pulse. In turn, these hydrological changes in riverine and riparian inputs will have both positive and negative effects on floodplain insect assemblages.

Tockner et al. (2010) viewed changes in the magnitude and phenology of floodplain inundation as stressors that could have profound effects on the productivity of floodplain systems, influencing organic decomposition as well as growth and emergence windows for aquatic insects. Depending on flood timing (see below), increases in floodplain connectivity will likely increase primary (Gallardo et al., 2023) and secondary production on the ARB floodplain relative to areas outside the ARB levees (Lu et al., 2024). A review by Batzer, (2012) indicated insect abundance and production generally increased with flooding duration in southeastern floodplain systems, with invertebrate taxa employing various strategies to exploit floodplain resources. Turić et al. (2015) reported similar positive effects of high flood-pulse magnitude and prolonged inundation on diversity and abundance of heteropteran and coleopteran taxa inhabiting the Danube River floodplain, although an increased hydroperiod was linked to reduced emergence of dipterans, hymenopterans, coleopterans and "homopteran"s in floodplain wetlands of the River Dalälven, Sweden (Vinnersten et al., 2014).

In the ARB, however, hypoxia resulting from floodplain organic matter decomposition during the annual flood pulse has been shown to be an overriding environmental influence on the distribution and abundance of floodplain fishes, zooplankton, and macroinvertebrates (e.g., Davidson et al., 1998; Fontenot et al., 2001; Kelso et al., 2003; Colon-Gaud et al., 2004; Kaller et al., 2011). Spatial and temporal severity of floodplain hypoxia and its effects on floodplain organisms is related to the magnitude and duration of floodplain inundation after water temperatures exceed 20C (Sabo et al., 1999b; Pasco et al., 2016; Kroes et al., 2022). Increased rainfall in the Mississippi River and subsequent flooding in the ARB will undoubtedly exacerbate the frequency and duration of floodplain hypoxia, with pervasive effects on aquatic insect physiology, reproduction, and growth (Galic et al., 2019) during years with extended or late (post-June) flood pulses. These effects would likely be less severe for most ARB hemipterans given

their surface respiration habits (Popham, 1960), but would be expected to significantly affect the activity, growth, emergence, reproduction, and survival of obligate water-breathing insect taxa (Saari et al., 2018). Depending on changes to flooding phenology and magnitude, changes in insect assemblage dynamics would be expected to strongly influence energy and nutrient flows of larval aquatic insects to predatory invertebrates and fishes (e.g., Corti et al., 1997; Farly et al., 2019) as well as subsidies of emerging adult insects to terrestrial food webs (Baxter et al., 2005; Wesner et al., 2020).

Differences in invertebrate assemblages between connected BC floodplain and unconnected LFP floodplain reinforce the importance of river-floodplain connectivity to aquatic invertebrates (Gallardo et al., 2008, 2014; Batzer et al., 2016). The two sampling locations are directly across the West Atchafalaya Basin Protection Levee from one another, separated by only 100 m of dry land for much of their 26-km border. Because of their close proximity across their extensive shared border, the disparity in taxa between BC and LFP (79 of 81 possible taxa in connected habitats compared with 51 of 81 possible taxa in unconnected habitats), higher richness in connected habitats, and greater densities of 21 of 24 taxa in connected habitats appeared to be a direct result of loss of connection to river flooding. These results reinforce the conclusions of Lu et al., (2024), who compared nearby leveed and unleveed floodplain wetlands and demonstrated that although adjacent, like BC and LFP in this study, aerial dispersal was insufficient to support invertebrate diversity. Moreover, as McDaniel and Batzer, (2019) suggested, habitats changes from lack of flooding over extended periods may result in disconnected habitats that are not attractive to aerial dispersal, even after reconnection. Wahl et al., (2021) reported aquatic invertebrates from the same regional taxonomic pool as this study exhibited poor aerial colonization in unfavorable habitats. Therefore, Hemiptera and other aquatic taxa are either using the disconnected LFP floodplain during brief temporal periods and leaving when conditions become unfavorable (Boersma and Lytle, 2014), or are simply not able to disperse to and colonize the LFP floodplain from nearby Bayou Chene and other aquatic habitats within the BC floodplain. Given the length of sampling in this study, temporary use is unlikely. Instead, in the more than 60 years since the construction of the West Atchafalaya Basin Protection Levee, the habitats in the two locations have potentially diverged in some meaningful aspect of suitability (McDaniel and Batzer, 2019) due to loss of river connectivity and riverine flooding (Lu et al., 2024) and replacement with precipitation-driven flooding (Tye, 1986). Although hemipterans have a deserved reputation as habitat generalists (Barbour et al., 1992; Guareschi et al., 2017; Taybi et al. 2021), aquatic connectivity, specifically aquatic dispersal pathways, does appear to be an important assemblage structuring factor.

Flood magnitude did not appear to influence the invertebrate assemblage, other than delay the shift from early to late-flood assemblages. Flood magnitude in this system has been shown to have an influence on water quality, habitat, and fishes and crayfish (Kaller et al., 2011; Alford and Walker, 2013; Bennett and Kozak, 2016). However, for macroinvertebrates, the flood of 2019, although substantially higher than average in magnitude and duration, was not sufficiently different in habitat or water quality impacts to elicit a major response from these invertebrates.

Long-term disconnection between disconnected floodplains and floodplain waterbodies, such as LFP impacts abiotic and biotic ecosystem components. In the case of LFP, a floodplain water body that was historically part of a larger lotic system during annual flooding, this study and Gallardo et al., (2023) documented substantial differences in aquatic organisms between the disconnected habitats and habitats still connected to the Atchafalaya River. Following levee construction, LFP effectively functions as a lentic system, which is similar to 11% of anthropogenically disconnected systems in the United States (Knox et al., 2022a), and has less macroinvertebrate diversity than the connected floodplain. Similar long-term disconnections converting lotic influenced systems into lentic systems have reduced fish (Miranda 2005; Dembkowski and Miranda 2014; Jiang et al. 2021) and macroinvertebrate diversity (Gallardo et al., 2008; Paillex et al., 2013; Obolewski et al., 2018) compared with connected habitats due to lack of accessibility to the organisms and differences in water chemistry and increased stratification, siltation patterns resulting in shallow and homogenous depth rather than heterogenous ridge and swale geomorphology, and macrophyte composition (e.g., Carlson Mazur et al., 2022; Knox et al., 2022b). Formerly wetted floodplain habitats that become permanently

dry and terrestrial, which are estimated at 30.6% of former floodplains in the United States (Knox et al., 2022a), clearly are lost habitats to aquatic biota. Abiotic components, including longitudinal, lateral, and vertical hydrologic exchange, nutrient exchange, and soil biochemistry, also change with disconnection from riverine flooding sources (Amoros and Bornette, 2002). When disconnection is coupled with drainage for agriculture (47% of disconnected floodplains in the United States) or development (11% of disconnected floodplains in the United States), groundwater storage and reduced vertical connectivity can greatly alter the former floodplain's soil biochemistry toward terrestrial (Knox et al., 2022b). Thus, whether driven to wetland and lentic habitats, like LFP, or to terrestrial agricultural and developed habitats, important abiotic processes are radically altered with consequences to biota.

Although improvement of biodiversity and ecosystem function through restoration of river-floodplain connectivity has met with mixed success depending on the system and biota under study (Stoltefaut et al., 2024 and citations therein), improvements in the magnitude and timing of floodplain inundation appear to be a fundamental component of successful restoration. Modification of the Mollicy Farms subunit adjacent to the Ouachita River in northern Louisiana through reforestation and improved connectivity has restored floodplain function, as has been reported other systems worldwide (Opperman et al., 2010). For the Atchafalaya River, fostering improved circulation and timing of floodplain inundation, and perhaps restoring inputs to historic floodplain areas like LFP that have been cut off from river inputs by guide levees (as in the Lower Mississippi River; Elsey-Quirk et al., 2019), could yield significant increases in biodiversity and ecosystem function in this highly modified floodplain river.

Biodiversity of aquatic insects and other invertebrates on river floodplains is enhanced by river-floodplain connectivity for many species (summarized in Gallardo et al., 2014). Moreover, connection histories (i.e., frequently or infrequently connected) can be reliable predictors of assemblage composition (Leigh and Sheldon, 2009). For many insects, results of this study are similar to other studies. For example, more Ephemeroptera and Trichoptera were directly or indirectly positively associated with connectivity, similar to Usseglio-Polatera and Tachet, (1994). Our finding that abundances of Coleoptera, Odonata, and Hemiptera were adversely impacted by loss of connectivity in the ARB differed from Bonada et al., (2006), Skern et al., (2010), and Gallardo et al., (2014); in the ARB, likely because these earlier studies did not examine habitats that were impacted by a complete loss of connectivity. Studies of restored river connectivity and riverine flooding (e.g., Obolewski et al., 2018; Pander et al., 2018) generally report increased number of insect species and density. Therefore, despite annual precipitation-driven flooding, lower richness and density in the disconnected LFP is consistent with river connection and restoration studies reinforcing the importance of flood pulses. In conclusion, this study demonstrates that Hemiptera and other arthropods in the ARB negatively responded to loss of river connectivity in this subtropical floodplain system, reinforcing the importance of maintaining and restoring river-floodplain connectivity to maintain aquatic biodiversity.

Acknowledgments We would like to acknowledge the Louisiana State University AgCenter, School of Renewable Natural Resources, and Entomology Department for their support of this work. This work was supported by the US Army Corp of Engineers (Grant Number MIPR W42HEM90940923). Partial support for this project came from the U.S. Department of Agriculture National Institute of Food and Agriculture, Hatch Program as Project Number LAB-94410 and McIntire-Stennis Cooperative Forestry Program as Project Number LAB-94335. This manuscript was approved for publication by the Director of the Louisiana Agricultural Experiment Station as MS 000-000-00000. There are many technicians and undergraduate students without which this could have never come to fruition—Tiffany Pasco, Mariah Taylor, Melanie Holton Bates, Chase Mathey, Bailey Jones King, and Sarah Catherine Buie. Also thank you to the watermen of the Atchafalaya basin and the staff at Lake Fausse Pointe State Park. Without them, we may have still been paddling in the swamp today.

Funding U.S. Army Corps of Engineers, MIPR W42HEM90940923, Michael D Kaller, National Institute of Food and Agriculture, LAB-94410, Michael D Kaller, LAB-94335, Michael D Kaller.

Data availability Data are available by contacting the authors.

Declarations

Competing interests The authors have no competing interests to declare that are relevant to the content of this article.

References

Alford, J. & M. Walker, 2013. Managing the flood pulse for optimal fisheries production in the Atchafalaya River basin, Louisiana (USA). River Research and Applications 29: 279–296.

Amoros, C. & G. Bornette, 2002. Connectivity and biocomplexity in waterbodies of riverine floodplains. Freshwater Biology 47: 761–776.

Anderson, M. J., K. E. Ellingsen & B. H. McArdle, 2006. Multivariate dispersion as a measure of beta diversity. Ecology Letters 9: 683–693.

Anderson, J. T., F. L. Zilli, L. Montalto, M. R. Marchese, M. McKinney, & Y.-L. Park. 2013. Sampling and processing aquatic and terrestrial invertebrates in wetlands. In Anderson, J. T. and C. A. Davis, (ed), Wetland Techniques, Volume 2 Organisms. Springer: 143–196.

Arrington, D. & K. Winemiller, 2006. Habitat affinity, the seasonal flood pulse, and community assembly in the littoral zone of a Neotropical floodplain river. Journal of the North American Benthological Society 25(1):250—262. https://doi.org/10.1899/0887-3593.

Balzan, M. V., 2012. Associations of dragonflies (Odonata) to habitat variables within the Maltese Islands: a spatio-temporal approach. Journal of Insect Science 12: 87.

Barbour, M. T., C. G. Graves, J. L. Plafkin, R. W. Wisseman & B. P. Bradley, 1992. Evaluation of EPA's rapid bioassessment benthic metrics: Metric redundancy and variability among reference stream sites. Environmental Toxicology and Chemistry 11: 437–449.

Bates, D., M. Maechler, B. Bolker & S. Walker, 2015. Fitting Linear Mixed-Effect Models Using lme4. Journal of Statistical Software 67: 1–48.

Battle, J. M. & T. B. Mihuc, 2000. Decomposition dynamics in the lower Atchafalaya, a large floodplain river. Hydrobiologia 418: 123–136.

Batzer, D. P., 2012. The seemingly intractable ecological responses of invertebrates in North American wetlands: a review. Wetlands 33: 1–15. https://doi.org/10.1007/s13157-012-0360-2.

Batzer, D. P. & A. Ruhí, 2013. Is there a core set of organisms that structure macroinvertebrate assemblages in freshwater wetlands? Freshwater Biology 88: 1647–1659.

Batzer, D. P., B. J. Palik & R. Buech, 2004. Relationships between environmental characteristics and macroinvertebrate communities in seasonal woodland ponds of Minnesota. Journal of the American Benthological Society 23(1): 50–68.

Batzer, D. P., G. B. Noe, L. Lee & M. Galatowitsch, 2018. A floodplain continuum for Atlantic Coast rivers of the southeastern U.S.: Predictable changes in floodplain biota along a river's length. Wetlands 38: 1–13.

Batzer, D. P., B. Gallardo, A. Boulton, & M. Whiles. 2016. Invertebrates of temperate-zone river floodplains. In Batzer, D. P. and D. Boix, editors. Invertebrates in Freshwater Wetlands. Springer: 451–492.

Baxter, C. V., K. D. Fausch & W. C. Saunders, 2005. Tangled webs: reciprocal flows of invertebrate prey link streams and riparian zones. Freshwater Biology 50: 201–220. https://doi.org/10.1111/j.1365-2427.2004.01328.x.

Bennett, M. G. & J. P. Kozak, 2016. Spatial and temporal pattens in fish community structure and abundance in the largest US river swamp, The Atchafalaya River floodplain. Ecology of Freshwater Fish 25: 577–589.

Besacier-Monbertrand, A.-L., A. Paillex & E. Castella, 2014. Short-term impacts of lateral hydrological connectivity restoration on aquatic macroinvertebrates. River Research and Applications 30: 557–570.

Boersma, K. S. & D. A. Lytle, 2014. Overland dispersal and drought-escape behavior in a flightless aquatic insect, Abedus herberti (Hemiptera: Belostomatidae). Southwestern Naturalist 59: 301–302.

Bonada, N., M. Rieradevall, N. Prat & V. H. Resh, 2006. Benthic macroinvertebrate assemblages and macrohabitat connectivity in Mediterranean-climate streams of northern California. Journal of the North American Benthological Society 25: 32–43.

Bonvillain, C. P., D. A. Rutherford, W. E. Kelso & C. E. Murphy, 2012. Biotic and abiotic influences on population characteristics of Procambarus clarkii in the Atchafalaya River Basin, Louisiana. Freshwater Crayfish 19: 125–136.

Bonvillain, C. P., D. A. Rutherford & W. E. Kelso, 2014. Effects of environmental hypoxia on population characteristics of red swamp crayfish *Procambarus clarkii* in the Atchafalaya River Basin, Louisiana. Hydrobiologia 743: 309–319.

Boulton, A. J. & L. N. Lloyd, 1991. Macroinvertebrate assemblages in floodplain habitats of the lower River Murray, South Australia. Regulated Rivers: Research and Management 6: 183–201.

Carlson Mazur, M. L., B. Smith, B. Bird, S. McMillan, M. Pyron & C. Hauswald, 2022. Hydrologic connectivity and land cover affect floodplain lake water quality, fish abundance, and fish diversity in floodplain lakes of the Wabash-White River basin. River Research and Applications 38: 160–172.

Chanut, P. C. M., T. Datry, C. Gabbud & C. T. Robinson, 2018. Direct and indirect effects of flood regime on macroinvertebrate assemblages in a floodplain riverscape. Ecohydrology. 12: 2095. https://doi.org/10.1002/eco.2095

Chessman, B. C. & L. Hardwick, 2014. Water regimes and macroinvertebrate assemblages in floodplain wetlands of the Murrmumbidgee River, Australia. Wetlands 34: 661–672.

Colon-Gaud, J. C. & W. E. Kelso, 2003. A suitcase trap for sampling macroinvertebrates in dense submerged aquatic vegetation. J. Kansas Entomol. Soc. 76: 119–123.

Colon-Gaud, J. C., W. E. Kelso & D. A. Rutherford, 2004. Spatial distribution of macroinvertebrates inhabiting hydrilla and coontail beds in the Atchafalaya Basin Louisiana. Journal of Aquatic Plant Management. 42: 85–91.

Corbet P. S. 1999. Dragonflies: Behaviour and Ecology of Odonata. Harley Books.

Corti, D., S. L. Kohler & R. E. Sparks, 1997. Effects of hydroperiod and predation on a Mississippi River floodplain invertebrate community. Oecologia 109: 154–165.

Davidson, N. L., Jr., W. E. Kelso & D. A. Rutherford, 1998. Relationships between environmental variables and the abundance of cladocerans and copepods in the

Atchafalaya River Basin. Hydrobiologia 379: 175–181. https://doi.org/10.1023/A:1003488332055.

Dembkowski, D.J. & L.E. Miranda, 2014. Environmental variables measured at multiple spatial scales exert uneven influence on fish assemblages of floodplain lakes. Hydrobiologia 721:129—144. https://doi.org/10.1007/s10750-013-1655-x

Dube, T., T. Pinceel, L. De Necker, V. Wepener, P. Lemmens & L. Brendonck, 2019. Lateral hydrological connectivity differentially affects the community characteristics of multiple groups of aquatic invertebrates in tropical pans in South Africa. Freshwater Biology 64: 2189–2203.

Elsey-Quirk, T., S. A. Graham, I. A. Mendelssoh, G. Snedden, J. W. Day, R. R. Twilley, G. Shaffer, L. A. Sharp, J. Pahl & R. R. Lane, 2019. Mississippi Reiver sediment diversions and coastal wetland sustainability: synthesis of responses to freshwater, sediment, and nutrient inputs. Estuarine, Coastal and Shelf Science 221: 170–183.

Epler, J. H. 2006. Identification Manual for the Aquatic and Semi-Aquatic Heteroptera of Florida. State of Florida Department of Environmental Protection Division of Water Resource Management.

Farly, L., C. Hudon, A. Cattaneo & G. Cabana, 2019. Seasonality of a floodplain subsidy to the fish community of a large temperate river. Ecosystems 22: 1823–1837. https://doi.org/10.1007/s10021-019-00374-w.

Fisher, J. C., W. E. Kelso & D. A. Rutherford, 2012. Macrophyte mediated predation on hydrilla-dwelling macroinvertebrates. Fundamental and Applied Limnology 181: 25–38.

Foltz, S. & S. Dodson, 2009. Aquatic Hemiptera community structure in stormwater retention ponds: a watershed land cover approach. Hydrobiologica 621: 49–62.

Fontenot, Q., D. Rutherford & W. Kelso, 2001. Effects of environmental hypoxia associated with the annual flood pulse on the distribution of larval sunfish and shad in the Atchafalaya River basin, Louisiana. Transactions of the American Fisheries Society 130: 107–116.

Ford, M. & J. A. Nyman, 2011. Preface: An overview of the Atchafalaya river. Hydrobiologia. 658: 1–5.

Galic, N., T. Hawkins & V. E. Forbes, 2019. Adverse impacts of hypoxia on aquatic invertebrates: a meta-analysis. Science of the Total Environment 652: 736–743. https://doi.org/10.1016/j.scitotenv.2018.10.225.

Gallardo, B., M. García, Á. Cabezas, E. González, M. González, C. Ciancarelli & F. A. Comín, 2008. Macroinvertebrate patterns along environmental gradients and hydrological connectivity within a regulated river-floodplain. Aquatic Science 70: 248–258.

Gallardo, B., S. Dolédec, A. Paillex, D. B. Arscott, F. Sheldon, F. Zilli, S. Mérigoux, E. Castella & F. A. Comín, 2014. Response of benthic macroinvertebrates to gradients in hydrologic connectivity: A comparison of temperate, subtropical, Mediterranean, and semiarid river floodplains. Freshwater Biology 59: 630–648.

Gallardo, K., M. Kaller, D. Rutherford & W. Kelso, 2023. The influence of the river flood pulse on floodplain periphyton assemblages. Wetlands 43: 1–16.

Golladay, S. W., B. W. Taylor & B. J. Palik, 1997. Invertebrate communities of forested wetlands in southwest Georgia, USA: Habitat use and influence of extended inundation. Wetlands 17: 383–393.

Gonsoulin, G. J., 1973. Seven Families of Aquatic and Semi-aquatic Hemiptera in Louisiana. Entomological News 84: 173–189.

Gordon, B. A., O. Dorothy & C. F. Lenhart, 2020. Nutrient retention in ecologically functional floodplains: a review. Water. https://doi.org/10.3390/w12102762.

Guareschi, S., A. Laini & M. M. Sánchez-Montoya, 2017. How do low-abundance taxa affect river biomonitoring? Exploring the response of different macroinvertebrate-based indices. Journal of Limnology 76: 9–20.

Heck, K. L., G. van Belle & D. Simberloff, 1975. Explicit calculation of the rarefaction diversity measurement and the determinant of sufficient sample size. Ecology 56: 1459–1461.

Hurlbert, S. H., 1971. The nonconcept of species diversity: A critique and alternative parameters. Ecology 52: 577–586.

Jennings, J. C., J. R. Bellmore, J. B. Armstrong & R. W. Wisseman, 2023. Effects of process-based floodplain restoration on aquatic macroinvertebrate production and community structure. River Research and Applications 39: 1709–1723.

Jiang, X., P. Zheng, L. Cao & B. Pan, 2021. Effects of long-term floodplain disconnection on multiple facets of lake fish biodiversity: Decline of alpha diversity leads to a regional differentiation through time. Science of the Total Environment 763: 144177.

Jones, J. I., J. F. Murphy, A. L. Collins, P. S. Naden & P. D. Armitage, 2012. The impact of fine sediment on macroinvertebrates. River Research and Applications. 28: 1055–1071.

Kaller, M. D., W. Kelso, B. T. Halloran & D. A. Rutherford, 2011. Effects of spatial scale on assessment of dissolved oxygen dynamics in the Atchafalaya River basin, Louisiana. Hydrobiologia 658: 7–15.

Kaller, M. D., R. Keim, B. Edwards, R. Harlan, T. Pasco, W. Kelso & D. A. Rutherford, 2015. Aquatic vegetation mediates the relationship between hydrologic connectivity and water quality in a managed floodplain. Hydrobiologica 760: 29–41.

Kelso, W. E., D. A. Rutherford & N. L. Davidson Jr., 2003. Diel vertical migration of cladocerans and copepods in the Atchafalaya River Basin floodplain. Journal of Freshwater Ecology 18: 259–268. https://doi.org/10.1080/02705060.2003.9664492.

Knox, R. L., R. R. Morrison & E. E. Wohl, 2022a. A river ran through it: Floodplains as America's newest relic landform. Science Advances 8: 25. https://doi.org/10.1126/sciadv.abo1082.

Knox, R. L., E. E. Wohl & R. R. Morrisson, 2022b. Levees don't protect, they disconnect: A critical review of how artificial levees impact floodplain functions. Science of the Total Environment 837: 155773.

Kroes, D. E., R. H. Day, M. D. Kaller, C. R. Demas, W. E. Kelso, T. E. Pasco, R. Harlan & S. Roberts, 2022. Hydrologic connectivity and residence time affect the sediment trapping efficiency and dissolved oxygen concentrations of the Atchafalaya River Basin. Water Resources Research. 58(11): e2021WR030731.

Lambou, V. W. & S. C. Hern, 1983. Transport of organic carbon in the Atchafalaya Basin, Louisiana. Hydrobiologia 98: 23–34.

Leigh, C. & F. Sheldon, 2009. Hydrological connectivity drives patterns of macroinvertebrate biodiversity in floodplain rivers of the Australian wet /dry tropics. Freshwater Biology 54: 549–571.

Lewis, J. W., S. E. Lytle & A. A. Tavakoly, 2023. Climate change projections of continental-scale streamflow across the Mississippi River Basin. Theoretical and Applied Climatology 151: 1012–1034. https://doi.org/10.1007/s00704-022-04243-w.

Lu, K., H. Wu, S. C. Jahnig & F. He, 2024. The impacts of reduced connectivity on multiple facets of aquatic insect diversity in floodplain wetlands. Science of the Total Environment 912: 169207.

Matheson, R., 1907. Belostoma eating a bird. Entomological News 18: 452.

McDaniel, C. H. & D. P. Batzer, 2019. Effects of river flow regulation beyond the channel: Multifaceted changes within a group of invertebrate floodplain specialists. Wetlands 39: 87–98.

McInerney, P. J., R. J. Stoffels, M. E. Shackelton & C. D. Davey, 2017. Flooding drives a macroinvertebrate biomass boom in ephemeral floodplain wetlands. Freshwater Science 36: 726–738.

Merritt, R. W., K. W. Cummins & M. B. Berg, 2008. An Introduction to the Aquatic Insects of North America, 4th ed. Kendall-Hunt Publishing, Dubuque, IA:

Mihuc, T. B., C. F. Bryan & L. T. Beck, 2000. Long-term comparison of water hyacinth (Eichornia crassipes) invertebrate assemblages in a sub-tropical river floodplain: 1975–1976 and 1994–1995. Internationale Vereinigung Für Theoretische und Angewandte Limnologie: Verhandlungen 27: 2535–2539.

Mihuc, T. B., J. M. Battle, J. R. Mihuc & C. F. Bryan, 2002. Macroinvertebrate communities in littoral regions of a large river-floodplain: Relationships with water quality and vegetation. Internationale Vereinigung Für Theoretische und Angewandte Limnologie: Verhandlungen 28: 1023–1026.

Miranda, L.E., 2005. Fish assemblages in oxbow lakes relative to connectivity with the Mississippi River. Transactions of the American Fisheries Society 134(6):1480—1489. https://doi.org/10.1577/T05-057.1

Munyai, L. F., L. Mugwedi, R. J. Wasseman, F. Dondofema & T. Dalu, 2023. Assessing fish and macroinvertebrates assemblages in relation to environmental variables in Makuleke floodplain pans: implications for diodiversity conservation. Wetlands 43: 93. https://doi.org/10.1007/s13157-023-01738-8.

Murdoch, W. W., M. A. Scott & P. Ebsworth, 1984. Effects of the General Predator, Notonecta (Hemiptera) Upon a Freshwater Community. The Journal of Animal Ecology 53: 791.

Neuwirth, R. 2022. RColorBrewer: ColorBrewer Palettes. R package version 1.1–3.

Obolewski, K., A. Strzelczak & K. Glińska-Lewczuk, 2014. Does hydrological connectivity affect the composition of macroinvertebrates on *Stratiotes aloides* L. in oxbow lakes? Ecological Engineering 66: 72–81.

Obolewski, K., K. Glińska-Lewczuk & M. Bąkowska, 2018. From isolation to connectivity: The effect of floodplain lake restoration as habitats for macroinvertebrate communities. Aquatic Sciences 80: 4.

Oksanen, J., F. G. Blanchet, M. Friendly, R. Kindt, P. Lengendre, D. McGlinn, P. R. Minchin, R. O'Hara B., G. L. Simpson, P Solymos, M. H. H. Stevens, E. Szoecs, & H. Wagner. 2019. vegan: Community Ecology Package. R package version 2.5–6. Available at: http://CRAN.Rproject. org/package=vegan.

Opperman, J. J., R. Luster, B. A. McKenney, M. Roberts & A. W. Meadows, 2010. Ecologically functional floodplains: connectivity, flow regime, and scale. Journal of the American Water Resources Association 46(2): 211–226.

Paillex, A., E. Castella & G. Carron, 2007. Aquatic macroinvertebrate response along a gradient of lateral connectivity in river floodplain channels. Journal of the North American Benthological Society 26: 779–796.

Paillex, A., S. Dolédec, E. Castella, S. Mérigoux & D. C. Aldridge, 2013. Functional diversity in a large river floodplain: anticipating the response of native and alien macroinvertebrates to the restoration of hydrological connectivity. Journal of Applied Ecology 50: 97–106.

Paillex, A., E. Castella, P. S. E. Zu Ermgassen & D. Aldridge, 2015. Testing predictions of changes in alien and native macroinvertebrate communities and their interaction after the restoration of a large river floodplain (French Rhône). Freshwater Biology 60: 1162–1175.

Pander, J., M. Mueller & J. Geist, 2018. Habitat diversity and connectivity govern the conservation value of restored aquatic floodplain habitats. Biological Conservation. 217: 1–10.

Papacek, M., 2001. Small aquatic and ripicolous bugs (Heteroptera: Nepomorpha) as predators and prey: the question of economic importance. European Journal of Entomology 98: 1–12.

Pasco, T. E., M. D. Kaller, R. Harlan, W. E. Kelso, D. A. Rutherford & S. Roberts, 2016. Predicting floodplain hypoxia in the Atchafalaya River, Louisiana, USA, a large, regulated southern floodplain river system. River Research and Applications 32: 845–855.

Peckarsky, B. L., P. R. Fraissinet, M. A. Penton, & D. J. Conklin Jr. 1990. Freshwater Macroinvertebrates of Northeastern North America. Comstock Publishing Associates.

Pederson, T., 2024. patchwork: The composer of plots. R Package Version 1: 3.

Piazza, B., 2014. The Atchafalaya River basin history and ecology of an American wetland. Texas A&M University Press.

Poi de Neiff, A. & R. Carignan, 1997. Macroinvertebrates on *Eichhornia crassipes* roots in two lakes of the Paraná River floodplain. Hydrobiologia 345: 185–196.

Polhemus, J. T., 2008. Aquatic and semiaquatic Hemiptera. In Merritt, R. W., K. W. Cummins & M. B. Berg (eds), An Introduction to the Aquatic Insects of North America 4th ed. Kendall-Hunt Publishing, Dubuque, IA: 385–424.

Pollard, J. E., S. M. Melancon & L. S. Blakey, 1983. Importance of bottomland hardwoods to crawfish and fish in the Henderson Lake area, Atchafalaya Basin, Louisiana. Wetlands 3: 1–25.

Popham, E. J., 1960. On the respiration of aquatic Hemiptera Heteroptera with special reference to the Corixidae. Proceedings of the Zoological Society of London 135: 209–242. https://doi.org/10.1111/j.1469-7998.1960.tb05842.x.

Quinn, G. P., T. J. Hillman & R. Cook, 2000. The response of macroinvertebrates to inundation in floodplain wetlands: a possible effect of river regulation? Regulated Rivers: Research and Management 16: 469–477.

R Core Team, 2023. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.

Roberston, D. M. & D. A. Saad, 2021. Nitrogen and phosphorus sources and delivery from the Mississippi/Atchafalaya River Basin: an update using 2012 SPARROW models. Journal of the American Water Resources Association 57(3): 406–429.

Ruhí, A. & D. P. Batzer, 2014. Assessing congruence and surrogacy among wetland macroinvertebrate taxa towards efficiently measuring biodiversity. Wetlands 34: 1061–1071.

Saari, G. N., Z. Wang & B. W. Brooks, 2018. Revisiting inland hypoxia: diverse exceedances of dissolved oxygen thresholds for freshwater aquatic life. Environmental Science Pollution Research 25: 3139–3150. https://doi.org/10.1007/s11356-017-8908-6.

Sabo, M. J., C. F. Bryan, W. E. Kelso & D. A. Rutherford, 1999a. Hydrology and aquatic habitat characteristics of a riverine swamp: I. Influence of flow on water temperature and chemistry. Regulated Rivers 15: 505–523.

Sabo, M. J., C. F. Bryan, W. E. Kelso & D. A. Rutherford, 1999b. Hydrology and aquatic habitat characteristics of a riverine swamp: II. Hydrology and the occurrence of chronic hypoxia. Regulated Rivers 15: 525–544.

Sarker, D. 2008. Lattice: Multivariate Data Visualization with R. Springer-Verlag, New York, New York.

Secor, D. H., J. M. Dean & J. Hansbarger, 1992. Modification of the quatrefoil light trap for use in hatchery ponds. The Progressive Fish Culturist 54: 202–205.

Shuman, T. C., P. C. Smiley, R. B. Gillespie & J. M. Gonzalez, 2020. Influence of physical and chemical characteristics of sediment on macroinvertebrate communities in agricultural headwater streams. Water 12(2976): 1–17. https://doi.org/10.3390/w12112976.

Skern, M., I. Zweimueller & F. Schiemer, 2010. Aquatic Heteroptera as indicators for terrestrialisation of floodplain habitats. Limnologica 40: 241–250.

Spieles, D. J. & W. J. Mitsch, 2000. Macroinvertebrate community structure in high- and low-nutrient constructed wetlands. Wetlands 20(4): 716–729.

Starr, S. M., J. P. Benstead & R. A. Sponseller, 2014. Spatial and temporal organization of macroinvertebrate assemblages in a lowland floodplain ecosystem. Landscape Ecology 29: 1017–1031.

Stoffels, R. J., P. Humphries, N. R. Bond & A. E. Price, 2022. Fragmentation of lateral connectivity and fish population dynamics in large rivers. Fish and Fisheries 23: 680–696.

Stoltefaut, T., P. J. Haubrock, E. A. R. Welti, N. J. Baker & P. Hauuse, 2024. A long-term case study indicates improvements in floodplain biodiversity after river restoration. Ecological Engineering 198: 107143.

Taybi, A. F., Y. Mabrouki, H. Bozdoĝan & A. Millán, 2021. Are aquatic Hemiptera good indicators of environmental river conditions? Aquatic Ecology 55: 791–806.

Thorp, J. H. & D. C. Rogers, 2011. True Bugs: Insect Order Hemiptera, Field Guide to Freshwater Invertebrates of North America Academic Press, NY: 205–212.

Tockner, K., M. Pusch, D. Borchardt, & M.S. Lorang, 2010. Multiple stressors in coupled river-floodplain ecosystems. Freshwater Biology 55(1): 135—151. https://doi.org/10.1111/j.1365-2427.2009.02371.x

Tronstad, L. M., B. P. Tronstad & A. C. Benke, 2007. Aerial colonization and growth: Rapid invertebrate responses to temporary aquatic habitats in a river floodplain. Journal of the North American Benthological Society 26: 460–471.

Trumbo, B. A., M. D. Kaller, A. R. Harlan, T. E. Pasco, W. E. Kelso & D. A. Rutherford, 2016. Effectiveness of point versus continuous electrofishing for fish assemblage assessment in shallow, turbid aquatic habitats. North American Journal of Fisheries Management 36: 398–406.

Turić, N., M. Temunovic, A. Radovic, G. Vignjevic, M. Sudaric Bogojevis & E. Merdic, 2015. Flood pulses drive the temporal dynamics of assemblages of aquatic insects (Heteroptera and Coleoptera) in a temperate floodplain. Freshwater Biology 60: 2051–2065.

Turner, R. E., 2022. Variability in the discharge of the Mississippi River and tributaries from 1817 to 2020. PLoS ONE. https://doi.org/10.1371/journal.pone.0276513.

Tye, R. S., 1986. Non-marine Atchafalaya Deltas: Processes and Products of Interdistributary Basin Alluviation, Louisiana State University, Ph.D. Disseratation, South-Central Louisiana:

Usseglio-Polatera, P. & H. Tachet, 1994. Theoretical habitat templets, species traits, and species richness: Plecoptera and Ephemeroptera in the Upper Rhone River and its floodplain. Freshwater Biology 31: 357–375.

Vargas-Lopez, I., W. Kelso, C. Bonvillain, R. Keim & M. D. Kaller, 2020. Influence of water quality, local knowledge and river floodplain connectivity on commercial wild crayfish harvesting in the Atchafalaya River basin. Fisheries Management and Ecology 27: 417–428.

Vinnersten, T. Z. P., Ö. Östman, M. L. Schäfer & J. O. Lundström, 2014. Insect emergence in relation to floods in the wet meadows and swamps in the River Dalälven floodplain. Bulletin of Entomological Research. https://doi.org/10.1017/S0007485314000078.

Wahl, C. F., R. Diaz & M. D. Kaller, 2021. Invasive floating fern limits aerial colonization and alters community structure of aquatic insects. Wetlands 41: 60. https://doi.org/10.1007/s13157-021-01457-y.

Wang, Y. I., U. Naumann, S. Wright & D. I. Warton, 2012. mvabound-an R package for model-based analysis of multivariate abundance data. Methods in Ecology and Evolution 3: 471–474.

Wesner, J. S., D. L. Swanson, M. D. Dixon, D. A. Soluk, D. J. Quist, L. A. Yager, J. W. Warmbold, E. Oddy & T. C. Seidel, 2020. Loss of potential aquatic-terrestrial subsidies along the Missouri River Floodplain. Ecosystems 23: 111–123. https://doi.org/10.1007/s10021-019-00391-9.

Wickham, H. 2016. ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag, New York, New York.
Willbanks, K. A. & D. L. Mullis, 2022. Flooding effects on aquatic invertebrates in oxbow lakes of a southeastern USA river floodplain. Hydrobiologia 849: 1495–1510.
Wilson, C. A., 1958. Aquatic and semi-aquatic Hemiptera of Mississippi. Tulane Studies in Zoology and Botony 6: 115–170.
Wissinger, S. A., A. J. Klemmer, A. Braccia, B. M. Bush & D. M. Batzer, 2021. Relationships between macroinvertebrates and detritus in freshwater wetlands. Freshwater Science. https://doi.org/10.1086/717487.
Zilli, F. L., L. Montalto & M. R. Marchese, 2008. Benthic invertebrate assemblages and functional feeding groups in the Paraná River floodplain (Argentina). Limnologica 38: 159–171.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.